

MicroNiti (सूक्ष्म-नीति)

AI-Driven Hyper-Local Advisory & Financial Structuring Platform for Rural Micro-Entrepreneurs

ORGANIZATION / TEAM

ORIGIN LABS-SJ

REPOSITORY

ORIGIN-LABS-SJ/SIH-2026

TARGET GROUPRural MSMEs, Artisans,
OBC/SC/ST**AUDIT DATE**

September 2026

1. Executive Summary & Problem Analysis (SIH26091 Synopsis)

MicroNiti is an AI-driven digital assistant designed to empower rural micro-entrepreneurs (small shopkeepers, artisans, local manufacturers, and agri-business owners) with localized business decision support and automated financial structuring. By delivering voice-first interactions in regional dialects, offline-capable micro-accounting, and tailored credit-readiness scoring, the platform closes the financial literacy gap and connects underserved enterprises to formal credit channels and hyper-local supply chains.

4 Core Problems Stated in Synopsis

- **Financial Exclusion:** Over 80% of rural micro-enterprises operate in informal cash economies, lacking structured financial records required by banks.
- **Language & Literacy Barriers:** Conventional business software relies heavily on complex English text interfaces, alienating rural users.
- **Lack of Local Market Intelligence:** Entrepreneurs make procurement/pricing decisions intuitively without visibility into wholesale mandi prices or demand trends.
- **Inaccessible Advisory & Middlemen:** Formal financial planning, tax guidance, and subsidy navigation remain inaccessible, leaving users vulnerable to predatory intermediaries.

5 Proposed Solutions & Practical Impacts

- **Voice-First Dialect Assistant:** Hands-free natural voice interaction in 10 regional dialects for low-literacy business guidance.
- **Offline Smart Ledger (Khata):** Localized, offline-first digital record keeping replacing paper notebooks with structured, tamper-proof income/expense logs.
- **Hyper-Local Market Intelligence:** Real-time tracking of nearby wholesale commodity rates, demand spikes, and competitor density.
- **Alternative Credit Scoring Engine:** ML evaluation based on cash flow consistency, utility bills, and supplier payments for MFIs and banks.
- **Government Scheme Matcher:** Automated eligibility mapping to MoSJE (NSFDC, NBCFDC), PMEGP, Vishwakarma, and MUDRA with direct subsidy linking.

2. Side-by-Side Comparison: Current Project vs. SIH Synopsis

CORE MODULE	SIH SYNOPSIS REQUIREMENT & PRACTICAL IMPACT	WHAT WE HAVE MADE TILL NOW (CURRENT STATUS)	PROGRESS	WHAT WE HAVE TO MAKE (REMAINING WORK)
1. Voice-First Dialect Assistant	NLP supporting regional languages and local dialects via speech-to-text; enables low-literacy users to manage inventory, log sales, and seek advice hands-free.	• Multilingual AI engine in <code>ai_service.py</code> supporting 10 languages & dialects (Marwari, Hindi, Gujarati, Marathi, Bengali, Punjabi, Tamil, Telugu, Kannada, English). • Google Gemini 2.0/1.5 Flash + local heuristic fallback. • Web Speech API microphone dictation button in UI.	80% DONE	• Voice-to-Ledger Action Parser: Enable hands-free speech input to directly log transactions (e.g., "<i>Aaj 600 rupaye ka tel becha</i>"). • Integrated Audio Speech Synthesis (TTS playback) for illiterate entrepreneurs.
2. Offline Smart Ledger (Khata)	Localized, offline-first digital record keeping with automatic cloud sync when connectivity returns. Replaces paper notebooks with structured, tamper-proof logging.	• High-precision capital distribution model (40% Stock, 25% Setup, 10% Marketing, 25% Cash Cushion). • Loan calculator and quarterly/monthly repayment planner. • Interactive Carousel & Bank Lending Modal in <code>main index.html</code>.	30% DONE	• Daily Transaction Ledger UI: Income, Expense, and Customer Credit/Udhar recording. • Offline-First Storage: IndexedDB / LocalStorage caching so it functions with zero internet. • Auto Cloud Sync Engine: Queued background synchronization to <code>/api/ledger/sync</code>.
3. Hyper-Local Market Intelligence	Real-time tracking of nearby wholesale prices, local demand spikes, and regional market trends to optimize procurement pricing and stock high-demand items.	• Reverse-geocoding via BigDataCloud & OpenStreetMap Nominatim with nearest Indian city fallback. • Interactive Leaflet satellite map picker. • Catchment competitor estimation and pedestrian footfall indicators in Results Dashboard.	50% DONE	• Wholesale Mandi Commodity Tracker: Real-time/cached wholesale rates for essential rural goods (grains, pulses, oil, textiles). • Local Demand Surge Radar: Seasonal/festival demand spike alerts and procurement timing suggestions.
4. Alternative Credit Scoring Engine	ML evaluation based on cash flow consistency, supplier metrics, and utility payments; generates a verifiable digital financial profile for MFIs and banks.	• Interactive CIBIL Simulator (300-900 dial gauge) displaying approval odds, interest rebate (-0.50%), and margin money in <code>CibilScore.jsx</code> & <code>index.html</code>.	55% DONE	• Non-Bureau ML Algorithm: Score based on Cash-Flow Regularity (35%), Supplier Discipline (25%), Utility Bills (20%), Digital UPI Ratio (20%). • Bank/MFI-Ready Digital Dossier: Downloadable verifiable credit certificate with QR code for loan officers.

CORE MODULE	SIH SYNOPSIS REQUIREMENT & PRACTICAL IMPACT	WHAT WE HAVE MADE TILL NOW (CURRENT STATUS)	PROGRESS	WHAT WE HAVE TO MAKE (REMAINING WORK)
5. Government Scheme Matcher	Automated engine mapping enterprise profiles to relevant central/state subsidies and loan schemes without intermediary exploitation.	• Consumer Favorability Scoring: Ranks schemes by maximum financial advantage. • Features NSFDC / NBCFDC (Ministry of Social Justice 6% concessional loans), PM Vishwakarma (5% + ₹15k grant), PMEGP (up to 35% subsidy), Stand-Up India, SVANidhi, MUDRA.	95% DONE	• State-specific rural livelihood schemes (e.g. NRLM / Aajeevika). • One-click checklist for required documentation and bank-readiness verification.
6. Tech Stack & Infrastructure	• Frontend: Flutter / PWA + Speech-to-Text. • Backend: Python (FastAPI) + Node.js. • AI: Meta Llama / Fine-tuned LLM + Bhashini API. • DB: PostgreSQL + PostGIS + SQLite.	• Frontend: Standalone Responsive Web App (PWA-ready) + React component library. • Backend: Python FastAPI (v3.1.0) with Uvicorn, CORS, Swagger at /docs. • AI: Google Gemini 2.0/1.5 Flash REST API + heuristic dialect translation. • DB: SQLite (khata.db) with spatial reverse-geocoding.	90% DONE	• Add Bhashini API translation pipeline references. • Expand SQLite schema to support offline sync transaction queues and commodity price caching.
7. Security & Identity Verification	AES-256 encryption at rest, TLS 1.3 in transit, backed by Aadhaar e-KYC integration options for bank verification.	• Google Identity Services (GIS) OAuth integration. • Phone OTP authentication with SQLite session persistence. • Salted SHA-256 password hashing.	70% DONE	• Aadhaar e-KYC Simulator: Instant mock OTP verification badge ("Aadhaar e-KYC Verified for Priority Banking") to embed in the MFI financial profile.

3. Multi-Phase Implementation Roadmap

To bridge the gap between our current codebase and 100% compliance with the SIH26091 synopsis, the project follows a structured 5-phase engineering roadmap:

Phase 1: Project Identity, Governance & Repository Setup

✓ COMPLETED

- Rebranded platform identity to **MicroNiti (SIH26091)** under the **Ministry of Social Justice and Empowerment**.
- Established GitHub Organization **`ORIGIN-LABS-SJ`** with private repository **ORIGIN-LABS-SJ/SIH-2026**.
- Configured branch protection rulesets (PR required, approvals enforced, owner-exclusive direct push).
- Created automated 1-click synchronization tool **sync_github.bat** for instantaneous remote updates.
- Designated and preserved the verified 410KB frontend file with Carousel & Bank Lending Options.

Phase 2: Offline Smart Ledger (Daily Digital Khata) Module

🚀 CURRENT FOCUS

- **Daily Transaction Logging UI:** Quick-action buttons for daily Cash/UPI Sales (Income), Stock & Operational Costs (Expenses), and Customer Udhar (Credit Book).
- **Offline-First Persistence:** Implement IndexedDB/LocalStorage storage ensuring 100% functionality without internet in rural areas.
- **Cloud Synchronization Queue:** Background worker checking connectivity and synchronizing offline batches to backend `/api/ledger/sync`.
- **Visual Audit Trail:** Tamper-proof, structured ledger view replacing paper notebooks with downloadable monthly summaries.

Phase 3: Alternative ML Credit Scoring Engine & MFI Dossier

📄 PLANNED

- **Alternative Credit Scoring Model (300–900):** Evaluates non-bureau variables:
 - *Cash Flow Regularity (35% weight):* Predictability and cadence of daily sales.
 - *Supplier Repayment Discipline (25% weight):* Punctuality in settling wholesale vendor dues.
 - *Utility & Rent Timeliness (20% weight):* Consistency in paying shop electricity, rent, and recharge.
 - *Digital Transaction Ratio (20% weight):* Proportion of verifiable UPI transactions vs. cash.
- **Verifiable Digital Financial Profile:** One-click export of an official **"Bank & MFI Credit Dossier"** with QR-code verification, summary ratios, and loan eligibility recommendations.

Phase 4: Hyper-Local Market Intelligence & Mandi Price Tracker

📄 PLANNED

- **Wholesale Mandi Rates Radar:** Real-time and cached commodity price board tracking rural commodities (grains, edible oils, packaging, textiles).
- **Procurement Timing Optimization:** Actionable alerts suggesting optimal bulk procurement windows to maximize margin.
- **Local Demand Spike Indicator:** Calendar-driven and seasonal demand alerts (festivals, wedding seasons, harvest).

Phase 5: Hands-Free Voice Transaction Logging & Aadhaar e-KYC Verification

📄 PLANNED

- **Voice-to-Ledger Action Parsing:** Enables users to speak in their regional dialect (e.g., *"Aaj 500 रुपये ki bikri hui"*) to automatically record an income entry into the Smart Ledger hands-free.
- **Aadhaar e-KYC Verification Simulator:** Instant mock Aadhaar OTP verification producing a verified badge for institutional credit approval.
- **Audio Text-to-Speech (TTS):** Reads financial advice aloud in regional dialects for low-literacy users.

 **Alignment with Ministry of Social Justice and Empowerment (MoSJE) Mandate**

MicroNiti specifically addresses the target demographic of the Ministry: backward classes (OBC), Scheduled Castes (SC), Scheduled Tribes (ST), rural traditional craftspeople, and women micro-entrepreneurs. By embedding **NSFDC** and **NBCFDC** concessional credit schemes (fixed 6% interest) alongside **PM Vishwakarma** (5% interest + ₹15k toolkit grant) as priority recommendations, MicroNiti bridges the gap between underprivileged rural entrepreneurs and formal financial empowerment.